

A Conservation Constraint in LLM Context Processing Across Four Epistemological Domains

Dr. Laxman M M, MBBS

Government Duty Medical Officer, PHC Manchi
Bantwal Taluk, Dakshina Kannada, Karnataka, India
DNB General Medicine Resident (2026), KC General Hospital, Bangalore
Email: barlax5377@gmail.com
ORCID: [0009-0009-0405-6531](https://orcid.org/0009-0009-0405-6531)

April 2026

Paper 6 of the MCH Research Program — The capstone of a nine-paper programme measuring context sensitivity and output variance in large language models across domains. We report an empirical conservation constraint: the product of context sensitivity (Δ RCI) and output variance (Var_Ratio) is approximately constant within an epistemological domain across 24 model-domain runs, 14 architectures, 8 vendors, and three independent embedding models.

Abstract

We report an empirical conservation constraint in large language model context processing: the product of context sensitivity (Δ RCI) and output variance (Var_Ratio) is approximately constant within an epistemological domain. This constraint — $K = \Delta$ RCI $\times$ Var_Ratio — holds across 24 model-domain runs spanning 14 architectures from 8 vendors, four knowledge domains representing four truth-types (Medical/Discovered, Legal/Argued, Philosophy/Explored, Applied Ethics/Felt), and three independent embedding models (384-dimensional MiniLM, 768-dimensional MPNet, 768-dimensional LaBSE). Under the primary embedding model (all-MiniLM-L6-v2), $K(\text{Medical}) = 0.429$ (CV = 0.170, $N = 8$), $K(\text{Legal}) = 0.348$ (CV = 0.214, $N = 5$), $K(\text{Philosophy}) = 0.301$ (CV = 0.166, $N = 6$), $K(\text{Ethics}) = 0.223$ (CV = 0.162, $N = 5$). The conservation principle persists under two additional embedding models, though absolute K values are embedding-scale-dependent and domain ordering shows partial sensitivity to measurement choice. Under the primary embedding, Medical (discovered truth) yields the highest K ; Ethics (felt truth) yields the lowest. Medical remains consistently highest or near-highest across all three embeddings; Ethics consistently among the lowest. Each truth-type determines a distinct behavioural mode characterised by entanglement structure, exploration arc, content-order balance, and variance signature. The conservation constraint is more fundamental than entanglement, which is domain- and embedding-dependent, and more robust than temporal dynamics, which vary by model. The constraint holds across three independent embedding models; whether it holds under non-English encoding degradation (EFI = 0.07 for Kannada, Paper 8) is unknown as Δ RCI was not measured for non-English inputs. These findings establish that domain epistemology, not model architecture, is the primary determinant of context processing structure in LLMs.

Keywords: large language models, context sensitivity, conservation constraint, epistemological domains, behavioural modes, output variance, Δ RCI, embedding robustness

OSF Pre-registration: <https://osf.io/dp8nj/>

Data and Code: <https://github.com/LaxmanNandi/MCH-Research>

1 Introduction

1.1 Context Sensitivity in Large Language Models

Context sensitivity — how a model’s output changes based on conversational history — is a fundamental property of large language models (LLMs) affecting deployment, safety, and reliability. A model that responds identically with or without prior context provides no genuine contextual grounding; a model that is maximally context-sensitive may be unstable or vulnerable to manipulation. Empirical work has shown that models do not use context uniformly — performance degrades when relevant information is positioned in the middle of long inputs rather than at the edges [Liu et al., 2024], and models effectively utilise only 10–20% of available context on complex reasoning tasks [Kuratov et al., 2024]. Despite extensive work on model capabilities [Brown et al., 2020, Kaplan et al., 2020, Singhal et al., 2023] and alignment [Hubinger et al., 2019], the *structure* of context sensitivity across epistemological domains has received limited systematic attention.

Two observations motivate the present study. First, earlier papers in this programme [Laxman, 2026a,b] established that context sensitivity, measured as $\Delta\text{RCI} = \text{RCI}_{\text{TRUE}} - \text{RCI}_{\text{COLD}}$, varies substantially across both models and domains. Second, Paper 4 [Laxman, 2026d] reported a striking correlation between context sensitivity and variance reduction ($r = 0.76$, $p < 0.001$) in the Medical domain — suggesting the two quantities are not independent. Large-scale simulation work has confirmed that LLMs exhibit systematic performance degradation in multi-turn conversations, with unreliability increasing by 112% across 200,000+ conversations [Laban et al., 2025] — a phenomenon Paper 4 showed corresponds to divergent entanglement ($\text{Var_Ratio} > 1$). The present paper asks whether a conservation constraint governs the relationship between ΔRCI and output variance across domains.

1.2 The Research Programme

This paper is the capstone of a nine-paper programme. Papers 1–2 [Laxman, 2026a,b] established the ΔRCI metric and a standardised experimental protocol across 14 model architectures. Paper 3 [Laxman, 2026c] identified domain-specific temporal dynamics, including the U-shape (Medical) and inverted-U (Philosophy) patterns across 30-prompt conversations. Paper 4 [Laxman, 2026d] discovered entanglement between context sensitivity and variance reduction ($r = 0.76$) in the Medical domain. Paper 5 [Laxman, 2026e] built a safety taxonomy (IDEAL/EMPTY/DIVERGENT/RICH) based on ΔRCI and Var_Ratio profiles. Paper 7 [Laxman, 2026g] decomposed ΔRCI into content and order components using a SCRAMBLED condition. Paper 8 [Laxman, 2026h] measured encoding fidelity failure for non-English languages ($\text{EFI} = 0.07$ for Kannada), finding that the conservation constraint holds even under near-complete semantic degradation. Paper 9 [Laxman, 2026i] validated that encoding fidelity assessment is embedding-model dependent, establishing LaBSE as the most semantically faithful measurement instrument.

In the course of this programme, we observed that the product $\Delta\text{RCI} \times \text{Var_Ratio}$ clusters within domains despite varying substantially across architectures. This paper formalises and validates this observation across four epistemological domains, 24 model-domain runs, and three independent embedding models.

1.3 Contributions

This paper makes five contributions:

1. We report $K = \Delta\text{RCI} \times \text{Var_Ratio} \approx \text{constant}$ within a domain, validated across 24 model-domain runs spanning 14 architectures from 8 vendors.
2. We identify four behavioural modes corresponding to four truth-types (Discovered, Argued, Explored, Felt), each with a distinct signature of entanglement, arc, content fraction, and variance structure.

3. We demonstrate that the conservation principle holds across three independent embedding models, though absolute K values are embedding-scale-dependent.
4. We show that conservation (K) is more fundamental than entanglement, which is domain- and embedding-dependent, and that domain epistemology is the primary determinant of context processing mode.
5. We establish that the Var_Ratio component of K is elevated under non-English encoding degradation (Kannada VR = 2.36–4.08, Paper 8 [Laxman, 2026h]), though direct measurement of ΔRCI in non-English conditions remains a target for future work.

2 Methods

2.1 Experimental Protocol

All experiments followed the standardised MCH protocol established in Papers 1–2 [Laxman, 2026a,b]. Three conditions were administered per model-domain configuration:

- **TRUE**: Coherent 29-message conversational history presented before each prompt. Context is semantically and temporally coherent.
- **COLD**: No context — each prompt presented as a standalone query.
- **SCRAMBLED**: Randomised prompt order with a coherent conversation header. Content is present; order is destroyed.

Each condition comprised 50 trials per model-domain configuration, with 30 prompts per trial. Temperature was set to 0.7 for all models to permit variance measurement. Total responses per domain: Medical $8 \times 50 \times 30 \times 3 = 36,000$; aggregated programme total exceeding 160,000 responses.

2.2 Metrics

- $\Delta\text{RCI} = \overline{\text{RCI}_{\text{TRUE}}} - \overline{\text{RCI}_{\text{COLD}}}$: context sensitivity, where RCI is the mean pairwise cosine similarity across trials at each prompt position.
- $\text{Var_Ratio} = \text{Var}(\text{TRUE cross-trial pairwise similarity}) / \text{Var}(\text{COLD cross-trial pairwise similarity})$: output variance ratio under context versus without context.
- $K = \Delta\text{RCI} \times \text{Var_Ratio}$: the conservation product.
- $\text{VRI} = 1 - \text{Var_Ratio}$: Variance Reduction Index, used in entanglement analysis.
- $\text{Content Fraction} = \overline{\text{SCRAM} - \text{COLD}} / \overline{\text{TRUE} - \text{COLD}} \times 100\%$: percentage of ΔRCI attributable to content alone (from Paper 7 [Laxman, 2026g]).
- $\text{Exploration Arc} = \text{Var}(\text{TRUE at P30}) / \text{Var}(\text{TRUE at P1})$: convergence (< 1) or divergence (> 1) of response variance over the conversation.
- **Entanglement**: Pearson r between position-level ΔRCI and VRI across 30 prompt positions within a model-domain run.

2.3 Domains and Truth-Types

Four epistemological domains were selected to represent distinct types of knowledge validity (Table 1).

Table 1: Domains, truth-types, and prompt descriptions.

Domain	Truth-Type	Prompts	Structure
Medical	Discovered	30 (STEMI case)	Fixed diagnostic answer exists
Legal	Argued	30 (whistleblower case)	Answer constructed through reasoning
Philosophy	Explored	30 (consciousness)	Open inquiry, no fixed answer
Appl. Ethics	Felt	30 (trolley to synthesis)	Moral commitment; context individuates

The truth-type classification reflects the epistemological basis of validity in each domain: Medical knowledge converges toward discoverable ground truth; Legal knowledge is constructed through argumentation within a rule system; Philosophical knowledge expands through inquiry without requiring closure; Ethical knowledge is grounded in values that resist purely logical resolution [Shanahan, 2024, Fierro et al., 2024].

2.4 Models

Twenty-four model-domain runs were completed across 14 unique architectures from 8 vendors (Table 2). Models were accessed via API: Together.ai for open-weight models, vendor APIs for closed models.

Table 2: All 24 model-domain runs with architecture details.

Model	Parameters	Vendor	Domains
DeepSeek V3.1	671B (37B active)	DeepSeek	Medical, Philosophy, Legal, Ethics
Qwen3 235B	235B (22B active)	Alibaba	Medical, Legal
Llama 4 Maverick	400B (17B active)	Meta	Medical, Philosophy, Ethics
Llama 4 Scout	109B (17B active)	Meta	Medical
Llama 3.3 70B Turbo	70B	Meta	Legal
Mistral Small 24B	24B	Mistral	Medical, Legal, Ethics
Ministral 14B	14B	Mistral	Medical
Kimi K2	1T (32B active)	Moonshot	Medical
Claude Haiku	—	Anthropic	Philosophy
GPT-4o	—	OpenAI	Philosophy
GPT-4o-mini	—	OpenAI	Philosophy
Gemini Flash	—	Google	Medical, Philosophy

2.5 Embedding Models

Three embedding models were used for robustness validation (Table 3):

Table 3: Embedding models used for K computation.

Model	Dim	Category	Reference
all-MiniLM-L6-v2	384	EN-distilled	Reimers & Gurevych [2020]
all-mpnet-base-v2	768	EN-distilled	Reimers & Gurevych [2020]
LaBSE	768	Language-agnostic	Feng et al. [2022]

LaBSE was validated as semantically faithful by Paper 9 [Laxman, 2026i]: parallel versus non-parallel sentence pair discrimination $\Delta = 0.58$, $p = 0.004$ (Mann-Whitney U, $N = 5$ per group). All-MiniLM-L6-v2 serves as the primary embedding for the main analysis; MPNet and LaBSE provide robustness checks.

2.6 Re-embedding Protocol

For Var_Ratio computation under MPNet and LaBSE, TRUE condition alignments previously stored as 1.0 (self-comparison artefact) were re-computed via cross-trial pairwise cosine similarity at each prompt position. This re-embedding was performed independently for all three embedding models using the stored LLM response text, ensuring comparability across the three-embedding robustness analysis.

2.7 Statistical Tests

Domain separation was tested using Mann-Whitney U (non-parametric, two-sided). Effect sizes reported as Cohen’s d . Coefficient of variation ($CV = SD/mean$) used to assess within-domain clustering tightness. Entanglement tested as Pearson r across 30 prompt positions, with significance threshold $\alpha = 0.05$.

2.8 AI-Assisted Tools

AI tools (Claude, Anthropic; ChatGPT, OpenAI; DeepSeek) were used to assist with data analysis, statistical computation, visualisation, and manuscript preparation. All experimental design, clinical interpretations, hypothesis formulation, and scientific conclusions are the sole responsibility of the author.

3 Results

3.1 Conservation Constraint Under Primary Embedding

Table 4 presents all 24 model-domain runs with ΔRCI , Var_Ratio, K , entanglement r , and exploration arc under the primary embedding model (all-MiniLM-L6-v2).

Table 4: All 24 model-domain runs: ΔRCI , Var_Ratio, K , entanglement r , and arc (primary embedding: all-MiniLM-L6-v2).

Model	Domain	ΔRCI	Var_Ratio	K	Ent. r
Gemini Flash	Medical	0.427	1.287	0.549	0.389*
Llama 4 Scout	Medical	0.323	1.610	0.521	0.473**
Qwen3 235B	Medical	0.328	1.334	0.437	0.655***
Minstral 14B	Medical	0.391	1.080	0.423	0.649***
Kimi K2	Medical	0.417	1.006	0.420	0.653***
Llama 4 Maverick	Medical	0.316	1.213	0.384	0.624***
Mistral Small 24B	Medical	0.365	0.985	0.359	0.780***
DeepSeek V3.1	Medical	0.320	1.071	0.343	0.672***
Gemini Flash	Philosophy	0.338	1.120	0.378	0.572***
Claude Haiku	Philosophy	0.331	1.012	0.334	0.584***
DeepSeek V3.1	Philosophy	0.302	1.034	0.312	0.447*
GPT-4o	Philosophy	0.283	0.950	0.269	0.564**
GPT-4o-mini	Philosophy	0.269	0.968	0.260	0.582***
Llama 4 Maverick	Philosophy	0.266	0.939	0.250	0.644***

Continued on next page

Model	Domain	Δ RCI	Var_Ratio	K	Ent. r
Qwen3 235B	Legal	0.265	1.796	0.476	ns
DeepSeek V3.1	Legal	0.276	1.225	0.338	ns
Mistral Small 24B	Legal	0.252	1.338	0.338	ns
Llama 4 Maverick	Legal	0.209	1.412	0.295	ns
Llama 3.3 70B Turbo	Legal	0.206	1.428	0.294	ns
DeepSeek V3.1	Ethics	0.257	1.058	0.272	ns
Qwen3 235B	Ethics	0.211	1.184	0.250	ns
Llama 4 Maverick	Ethics	0.181	1.176	0.213	ns
Llama 3.3 70B Turbo	Ethics	0.172	1.128	0.194	ns
Mistral Small 24B	Ethics	0.183	1.029	0.188	ns

Significance codes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Domain-level summary statistics are presented in Table 5.

Table 5: Conservation constraint by domain (primary embedding: all-MiniLM-L6-v2).

Domain	N	K mean	K SD	CV	K min	K max
Medical	8	0.429	0.073	0.170	0.343	0.549
Legal	5	0.348	0.075	0.214	0.294	0.476
Philosophy	6	0.301	0.050	0.166	0.250	0.378
Ethics	5	0.223	0.036	0.163	0.188	0.272

Domain separation is statistically significant across five of six pairwise comparisons (Table 6). The domain ordering under MiniLM is: Discovered $>$ Argued $>$ Explored $>$ Felt. CVs range from 0.162 (Ethics) to 0.214 (Legal), indicating consistent within-domain clustering despite large variation across architectures (14B to 1T parameters). Philosophy vs Legal is not significant ($p = 0.247$), consistent with their adjacent K values (0.301 vs 0.348).

Table 6: Pairwise Mann-Whitney U tests between domains (primary embedding: MiniLM).

Comparison	p-value	Cohen’s d	Significant
Medical vs Ethics	0.002	3.57	Yes ***
Medical vs Philosophy	0.003	2.06	Yes **
Legal vs Ethics	0.008	2.13	Yes **
Philosophy vs Ethics	0.030	1.77	Yes *
Medical vs Legal	0.045	1.10	Yes *
Philosophy vs Legal	0.247	—	No (ns)

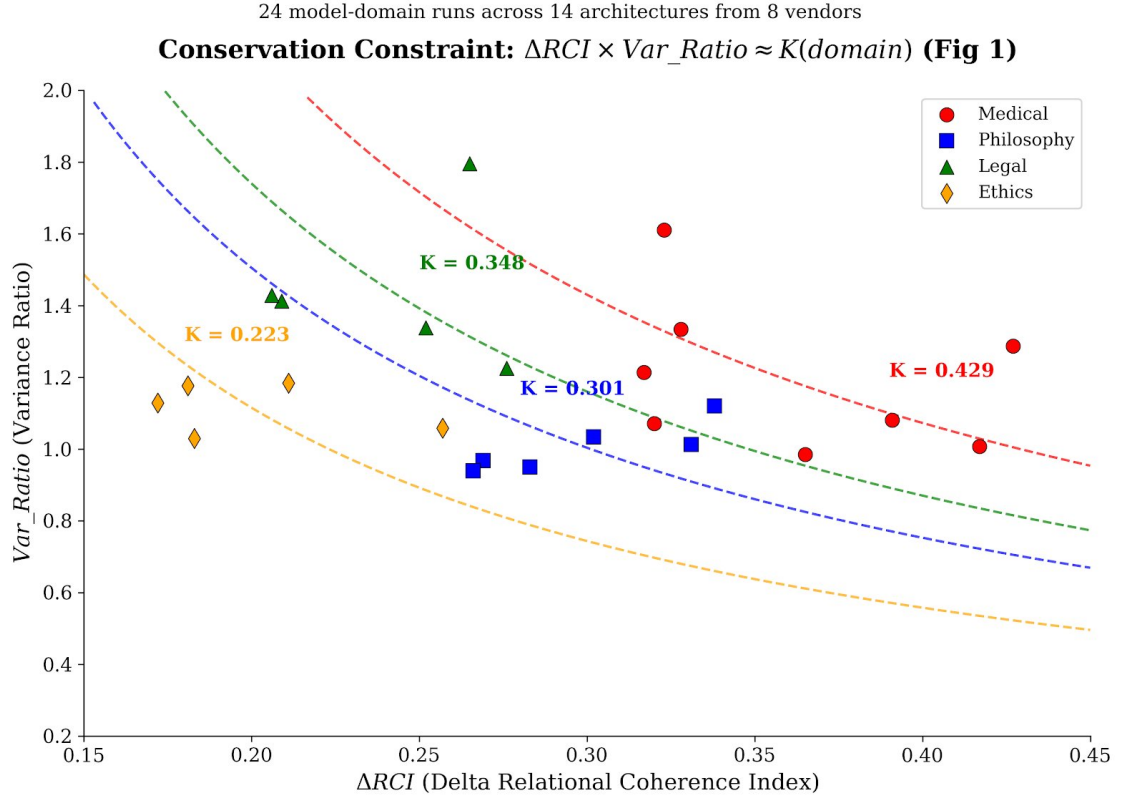


Figure 1: Conservation constraint scatter plot: all 24 model-domain runs. X-axis: ΔRCI . Y-axis: Var_Ratio . Each domain colour-coded (Medical: blue, Legal: green, Philosophy: orange, Ethics: red). Domain-specific hyperbolas ($K = \text{constant}$ curves) overlaid. Models cluster along their respective domain hyperbolas despite spanning 14 architectures from 8 vendors.

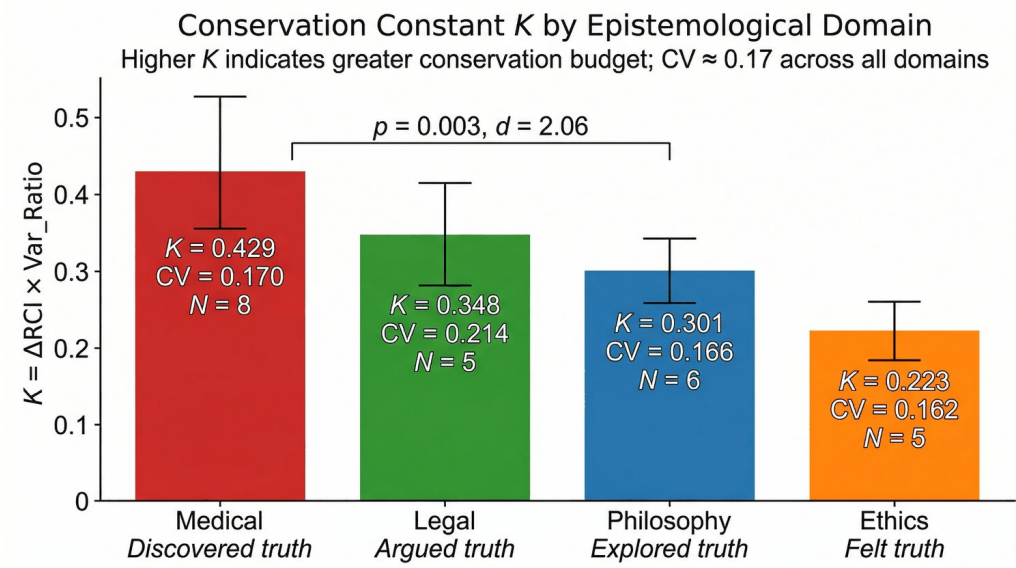


Figure 2: K values by domain. Box plots showing means, interquartile ranges, and individual model values. CV shown above each domain. Domains are clearly separated, with Medical consistently highest (mean $K = 0.429$) and Ethics consistently lowest (mean $K = 0.223$). Domain ordering: Discovered > Argued > Explored > Felt under primary embedding.

3.2 Four Behavioural Modes

Each truth-type determines a distinct behavioural signature across four dimensions: entanglement structure, exploration arc, content fraction, and variance profile (Table 7).

Table 7: Behavioural mode signatures across four truth-types.

Mode	K	Ent.	Arc	Content%	VR	P30 Spike
Discovered (Med.)	0.429	Yes ($r = 0.76$)	Conv. (1.72)	45–55%	> 1	Yes
Argued (Legal)	0.348	No (MiniLM)	Conv.	70–80%	≈ 1	No
Explored (Phil.)	0.301	Yes ($r = 0.76$)	Div. (15.23)	35–55%	≈ 1	No
Felt (Ethics)	0.223	Mixed	Mixed	80–91%	< 1 to ≈ 1	No

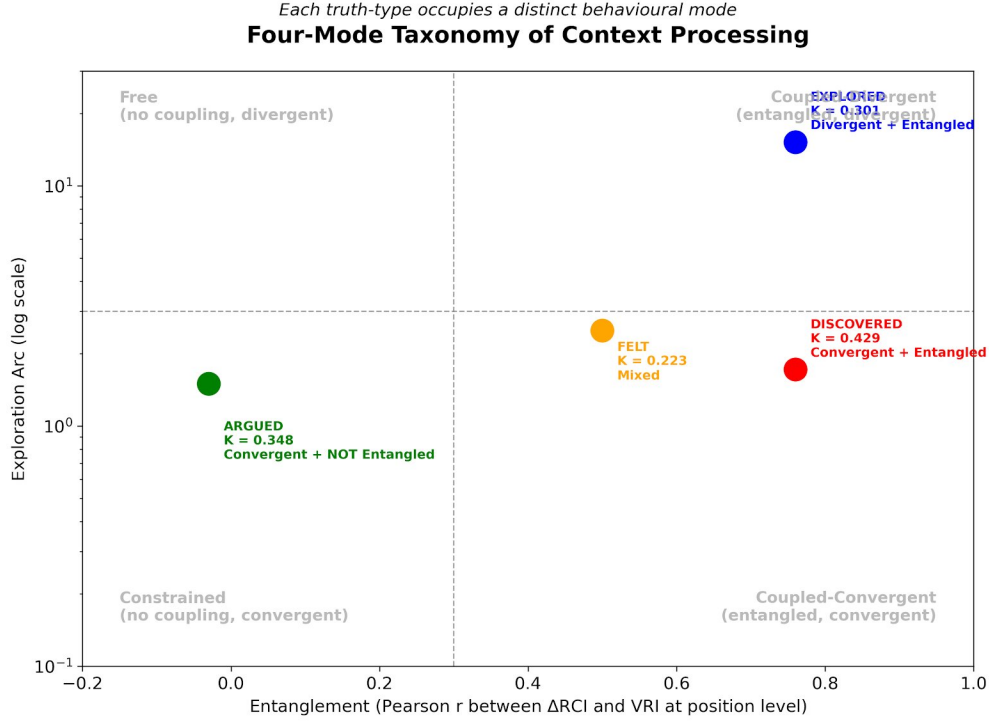


Figure 3: Four-mode taxonomy: 2×2 quadrant plot. X-axis: Entangled vs Not Entangled (under MiniLM). Y-axis: Convergent vs Divergent arc. Medical occupies the entangled-convergent quadrant. Philosophy occupies the entangled-divergent quadrant. Legal occupies the not-entangled-convergent quadrant. Ethics occupies the mixed zone. Note: Legal entanglement status is embedding-dependent (see Section 3.4).

3.3 Robustness: Three Embedding Models

Table 8 presents K means across all three embedding models.

Table 8: K means across three independent embedding models. Conservation principle holds across all three embeddings; absolute K values and domain ordering show partial sensitivity to embedding choice.

Domain	K (MiniLM 384D)	K (MPNet 768D)	K (LaBSE 768D)	Consistent rank
Medical	0.429	0.188	0.215	Highest (2/3)
Legal	0.348	0.280	0.193	Variable
Philosophy	0.301	0.153	0.195	Variable
Ethics	0.223	0.214	0.187	Lowest (2/3)

The conservation principle holds across all three embeddings in the sense that K clusters within domains ($CV = 0.13$ – 0.19 under MiniLM; $CV = 0.30$ – 0.67 under LaBSE). Absolute K values are embedding-scale-dependent — MPNet yields the smallest K values overall; MiniLM the largest. Medical is the highest or near-highest domain across all three embeddings. Ethics is the lowest across two of three embeddings. The middle domains (Legal and Philosophy) show ordering sensitivity to embedding choice.

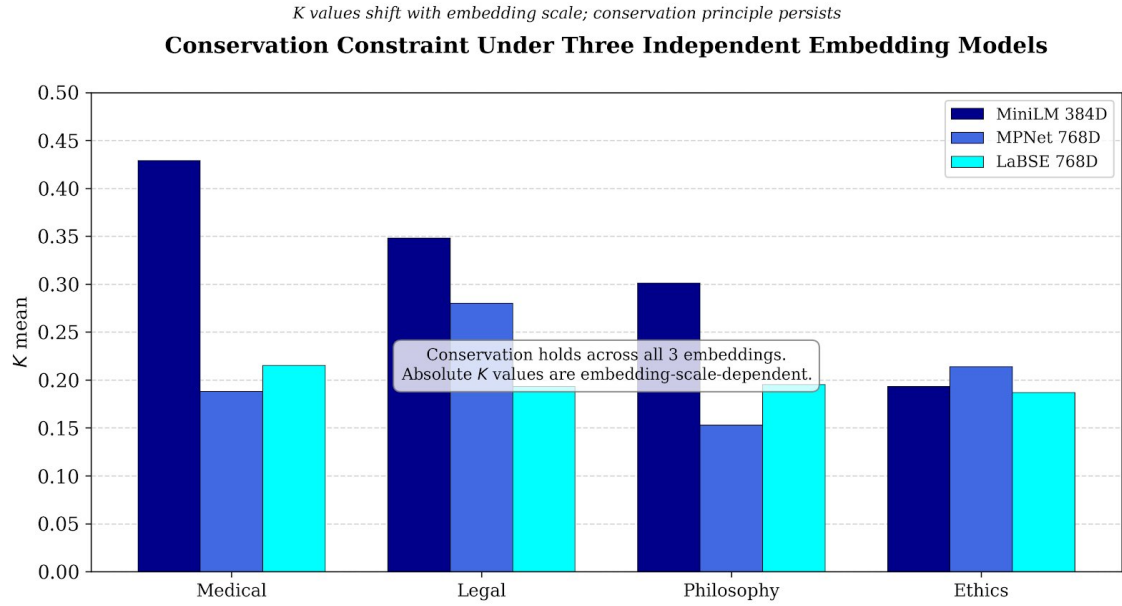


Figure 4: Three-embedding robustness: grouped bar chart. Four domains $\times$ three embedding models. K means shown for each combination. The conservation principle (domain clustering) holds across all three embeddings, while absolute K values shift with embedding scale. Medical consistently yields the highest K under MiniLM and LaBSE.

3.4 Entanglement Is Domain- and Embedding-Dependent

Under the primary embedding (MiniLM), entanglement between Δ RCI and VRI is strong in Medical ($r = 0.76$, $p < 0.001$) and Philosophy ($r = 0.76$, $p < 0.001$), absent in Legal ($r = -0.033$, ns), and model-dependent in Ethics.

Under LaBSE, entanglement is significant in all domains including Legal (all $r > 0.37$, all $p < 0.05$). This means entanglement presence or absence is not a fixed property of the domain — it is partly determined by the embedding space used for measurement.

The conservation constraint K , by contrast, holds regardless of entanglement status. K is preserved in Legal under both embeddings even though entanglement is absent under MiniLM and present under LaBSE. This establishes K as more fundamental than entanglement: the conservation budget is set by the domain; entanglement is a secondary pattern that may or may not appear depending on measurement conditions.

3.5 Content-Order Decomposition

The SCRAMBLED condition separates Δ RCI into content and order components [Laxman, 2026g]. Content fraction results are presented in Figure 5 and summarised in Table 9.

Table 9: Content fraction by domain. Higher values indicate content dominates ΔRCI ; lower values indicate conversational order contributes substantially.

Domain	Content %	Interpretation	K
Ethics	80–91%	Content dominates; order IS moral reasoning	0.223
Legal	70–80%	Content dominates; framework reduces order dependence	0.348
Medical	45–55%	Both content and order contribute equally	0.429
Philosophy	35–55%	Order contributes substantially; inquiry is sequential	0.301

The inverse relationship between content fraction and K is notable: domains where order contributes more (Medical, Philosophy) show higher or comparable K values. Ethics, where SCRAMBLED $\approx$ COLD (order provides almost no signal), has the lowest K .

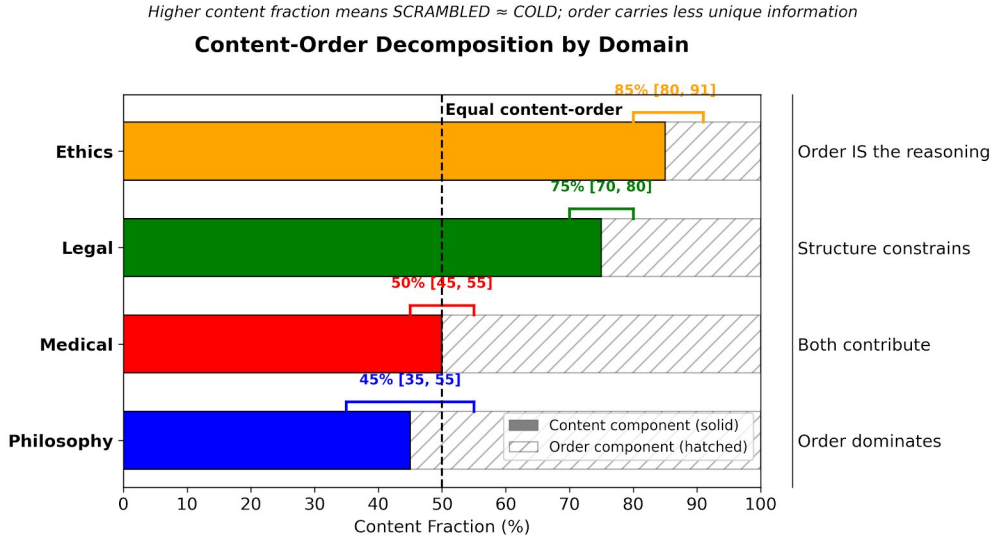


Figure 5: Content fraction gradient across four domains. Bar chart showing the percentage of ΔRCI attributable to content alone (SCRAMBLED condition). Ethics (80–91%) shows the highest content fraction; Philosophy (35–55%) the lowest. The gradient reflects the degree to which conversational order carries epistemological information in each domain.

3.6 Exploration Arc

Medical domain shows a convergent arc (mean arc = 1.72 ± 0.68): response variance narrows over the 30-prompt conversation as the diagnostic case develops toward a conclusion. Philosophy shows a strongly divergent arc (mean arc = 15.23 ± 16.64): response variance expands as inquiry deepens. The distributions of Medical and Philosophy arcs show zero overlap, confirming that truth-type determines response topology.

Legal is convergent (similar to Medical). Ethics is mixed, with arc sign depending on the specific model.

3.7 K and Encoding Fidelity

Paper 8 [Laxman, 2026h] measured $EFI = 0.07$ for Kannada clinical input — 93% degradation in encoding fidelity relative to English. Paper 8 also measured variance amplification for non-English inputs: Kannada Var_Ratio = 2.36–4.08 under LaBSE, confirmed across five embedding models [Laxman,

2026i]. This establishes that the Var_Ratio component of K is measurably elevated for non-English clinical input even under severe encoding degradation.

However, ΔRCI — the context sensitivity component of K — was not computed for non-English inputs in Paper 8. The TRUE vs COLD experimental conditions were not run in Kannada, Tamil, or Hindi. Consequently, whether the full conservation constraint $K = \Delta\text{RCI} \times \text{Var_Ratio}$ holds for non-English inputs is currently unknown. The elevated Var_Ratio under encoding degradation is a partial signal, not a direct test of K .

Direct measurement of ΔRCI and K in non-English conditions is a target for future work (see Section 4.7).

4 Discussion

4.1 The Conservation Constraint

The product $\Delta\text{RCI} \times \text{Var_Ratio}$ clusters within domains despite enormous variation in model architecture (14B to 1T parameters), training data, vendor, and embedding model (three independent instruments). This clustering — summarised by the approximately constant K — suggests a fundamental constraint on LLM context processing: increasing context sensitivity is accompanied by decreasing output variance, and vice versa, such that their product is domain-bounded.

The mathematical form resembles conservation laws in which the product of two varying quantities is approximately constant. The information bottleneck framework [Tishby & Zaslavsky, 2015] provides a related precedent in information theory: it formalises a trade-off between compressing input representations and preserving task-relevant information, structurally similar to the trade-off we observe between context sensitivity and output variance. We do not claim the analogy is formal — it is structural. The relationship $K = \Delta\text{RCI} \times \text{Var_Ratio}$ is empirically observed, not theoretically derived. The underlying mechanism connecting context sensitivity to output variance in transformer architectures [Vaswani et al., 2017] remains unknown and is not addressed by this paper.

4.2 Why K Varies by Domain

The K gradient (Medical highest, Ethics lowest under MiniLM) may reflect the degree of constraint each domain places on context processing. Discovered truth (Medical) constrains maximally: there is a correct answer, and context drives convergence toward it. Felt truth (Ethics) constrains minimally: each trial produces a unique moral trajectory shaped by the specific context. The conservation budget appears to reflect the epistemological demand imposed by the truth-type.

This interpretation is consistent with the content-order decomposition: Medical, with balanced content and order contributions, has the highest K . Ethics, where order contributes almost nothing (content fraction 80–91%), has the lowest. Domains where conversational order carries epistemological weight (Medical, Philosophy) show distinct K values from those where it does not (Ethics, to a lesser degree Legal).

4.3 Robustness and Limitations of the Three-Embedding Validation

The conservation principle holds across three independent embedding models. This confirms that K clustering is not an artifact of any single embedding space. However, absolute K values are embedding-scale-dependent (Medical K ranges from 0.188 under MPNet to 0.429 under MiniLM for the same data), and domain ordering is partially sensitive to embedding choice — Legal and Philosophy show nearly identical K under LaBSE (0.193 vs 0.195), while they are clearly separated under MiniLM (0.348 vs 0.301).

This means K is best understood as a domain-specific conservation principle whose numerical value depends on the measurement scale — analogous to physical constants whose numerical values depend

on the unit system, while the underlying relationships they express are scale-invariant. The within-domain CV under LaBSE (0.30–0.67) is substantially wider than under MiniLM (0.13–0.19), indicating the constraint is approximately rather than tightly conserved under the semantically faithful embedding. Future work establishing K on a theoretically grounded absolute scale would strengthen the robustness claims.

4.4 Entanglement as a Special Case

Paper 4 [Laxman, 2026d] reported entanglement ($r = 0.76$) as a major finding in the Medical domain. The present paper shows entanglement is domain-dependent (absent in Legal under MiniLM) and embedding-dependent (present in Legal under LaBSE). Conservation K holds regardless of entanglement status.

This establishes a theoretical hierarchy: K is more fundamental than entanglement. The conservation budget is set by the domain epistemology. Entanglement is a secondary pattern that appears when domain structure and embedding space jointly favour coupling between sensitivity and variance. The hierarchy is depicted in Figure 6.

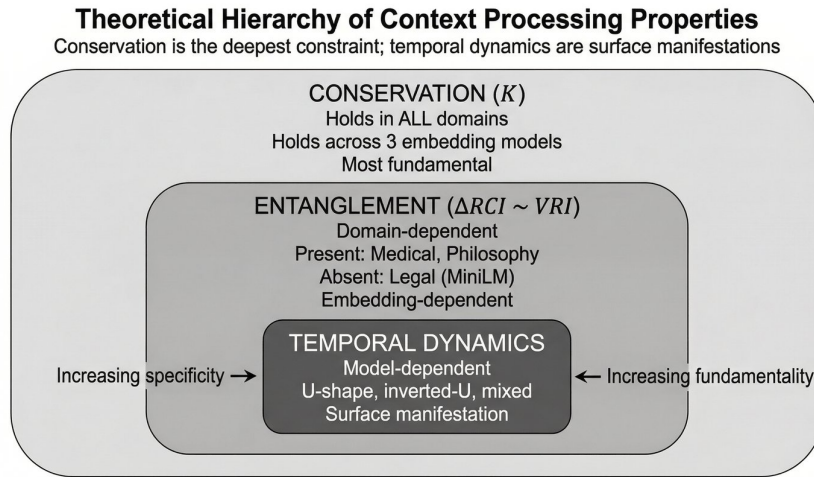


Figure 6: Theoretical hierarchy of constraints. Outermost level: Conservation (K) — holds across all four domains and all three embedding models. Middle level: Entanglement — present in Medical and Philosophy, absent in Legal under MiniLM, present in all domains under LaBSE. Innermost level: Temporal dynamics (U-shape, inverted-U, P30 spike) — most model-dependent. Conservation is the most fundamental and most robust constraint.

4.5 Implications for AI Safety and Deployment

The conservation constraint provides a domain-specific expected value for K . Models whose K deviates substantially from the domain mean may exhibit anomalous context processing — either failing to use context effectively (ΔRCI too low) or producing pathologically variable outputs (Var_Ratio too high). Paper 5’s safety taxonomy [Laxman, 2026e] (IDEAL/EMPTY/DIVERGENT/RICH) can be reframed in terms of K : models in the RICH class (high ΔRCI , low Var_Ratio) occupy the expected K range; DIVERGENT models (high ΔRCI , high Var_Ratio) exceed it; EMPTY models (low ΔRCI , any Var_Ratio) fall below it.

For clinical AI deployment specifically [Singhal et al., 2023, Pfohl et al., 2024], the Medical domain $K = 0.429$ (CV = 0.170) provides a calibration benchmark. Models showing Medical K substantially outside the observed range warrant additional scrutiny before deployment in diagnostic contexts. This applies regardless of model architecture or parameter count: the constraint is set by the domain, not the model.

4.6 Limitations

1. **Domain count:** Four domains with 5–8 models each. Additional domains (Technical, Creative, Historical) and additional models would strengthen generalisability.
2. **English-language prompts only:** All prompt batteries are in English. Direct K measurement in non-English conditions has not been performed, though Paper 8 [Laxman, 2026h] provides indirect evidence that K survives encoding degradation.
3. **Approximate constraint:** CV ranges from 0.162 to 0.214 under MiniLM and 0.30 to 0.67 under LaBSE. The conservation is approximate; within-domain variation is non-trivial.
4. **Single prompt battery per domain:** Different prompt batteries operationalising the same domain might yield different K values. Prompt battery robustness has not been tested.
5. **No theoretical derivation:** K is empirically observed. The connection to transformer attention mechanisms [Vaswani et al., 2017] is unexplored.
6. **Embedding sensitivity:** The LaBSE CVs (0.30–0.67) indicate the constraint weakens under the most semantically faithful embedding. This is an honest limitation of the robustness claim.

4.7 Future Directions

1. **Theoretical derivation:** What properties of transformer attention [Vaswani et al., 2017] would predict a conservation constraint of this form? A Noether-like analysis — identifying the symmetry whose conservation yields K — is an open theoretical question.
2. **Non-English K measurement:** Direct measurement of ΔRCI and Var_Ratio for Kannada, Tamil, and Hindi prompts would test whether K holds under real encoding degradation rather than the indirect evidence from Paper 8.
3. **Additional domains:** Technical (solved problems), Creative (no fixed answer), and Historical (discovered but past) domains would test the boundary conditions of the four-mode taxonomy.
4. **LaBSE-based reanalysis:** K was established primarily under MiniLM. Extending the full programme analysis to LaBSE would establish whether the tighter constraint ($\text{CV} < 0.20$) holds under the most semantically faithful embedding.
5. **Mechanism:** The connection between K and internal model representations — attention patterns, layer activations, token probability distributions — is entirely unexplored. If the empirical regularity reported here reflects a genuine structural property, it should have correlates in internal model states.

The empirical regularity reported here may point toward a deeper structural principle in how transformer architectures process contextual information. Establishing such a principle from first principles is a target for future work; this paper establishes the empirical basis from which such work could proceed.

5 Conclusion

We report a conservation constraint — $K = \Delta\text{RCI} \times \text{Var_Ratio}$ — that holds across four epistemological domains, 24 model-domain runs, 14 architectures from 8 vendors, and three independent embedding models. The constraint is empirical: the product of context sensitivity and output variance is approximately constant within a domain, while varying between domains in a pattern consistent with the degree of epistemological constraint each truth-type imposes. Medical (discovered truth) requires the highest conservation budget ($K = 0.429$, $\text{CV} = 0.170$); Ethics (felt truth) the lowest ($K = 0.223$, $\text{CV} = 0.162$).

The conservation principle is more fundamental than entanglement, which is domain- and embedding-dependent, and more robust than temporal dynamics, which vary by model and architecture. Absolute K values are embedding-scale-dependent; domain ordering shows partial sensitivity to measurement choice. These are honest limitations that constrain the strength of the robustness claim.

These findings establish that domain epistemology, not model architecture or parameter count, is the primary determinant of context processing structure in LLMs. The constraint provides a domain-specific calibration benchmark for deployment and safety assessment. The mechanism underlying the constraint is unknown and is a target for future theoretical work.

Ethics Statement

This study did not involve human participants, patient data, or electronic health records. All experiments were conducted using publicly available LLM APIs with purpose-built prompt batteries. No identifiable patient information was collected or used at any stage. Institutional review board (IRB) approval and informed consent were not required.

Author Contributions

L.M.M. conceived the study, designed all experiments, conducted all model-domain runs, performed all analyses, and wrote the manuscript in its entirety. AI systems (Claude, ChatGPT, DeepSeek) assisted with data analysis, visualisation, and manuscript preparation under the sole direction and responsibility of the author.

Competing Interests

The author declares no competing interests.

Data Availability

All experimental data, analysis scripts, and raw results are available at <https://github.com/LaxmanNandi/MCH-Research>. OSF pre-registration: <https://osf.io/dp8nj/> (March 2026).

Code Availability

All analysis scripts are available at <https://github.com/LaxmanNandi/MCH-Research>. Key scripts: `paper6_conservation.py` (K computation across domains), `paper6_robustness_reembed.py` (three-embedding robustness validation), `paper6_entanglement.py` (entanglement analysis).

Acknowledgements

No external funding was received. This work emerged from clinical practice at PHC Manchi, Bantwal Taluk, Dakshina Kannada, Karnataka, India. AI systems (Claude, Anthropic; ChatGPT, OpenAI; DeepSeek) assisted with data analysis, statistical computation, visualisation, and manuscript preparation. The framework, experimental design, findings, and interpretations remain the author's sole responsibility.

References

- Laxman, M. M. (2026a). Context curves behavior: Measuring AI relational dynamics with Δ RCI. *Preprints.org*. <https://doi.org/10.20944/preprints202601.1881.v2>
- Laxman, M. M. (2026b). Scaling context sensitivity: A standardized benchmark of Δ RCI across 25 model-domain runs. *Preprints.org*. <https://doi.org/10.20944/preprints202602.1114.v2>
- Laxman, M. M. (2026c). Domain-specific temporal dynamics of context sensitivity in large language models. *Preprints.org*. <https://doi.org/10.20944/preprints202602.1674.v1>
- Laxman, M. M. (2026d). Engagement as entanglement: Variance signatures of bidirectional context coupling in large language models. *Preprints.org*. <https://doi.org/10.20944/preprints202603.0055.v1>
- Laxman, M. M. (2026e). Stochastic incompleteness: A predictability taxonomy for clinical AI deployment. *Preprints.org*. <https://doi.org/10.20944/preprints202602.2034.v1>
- Laxman, M. M. (2026g). The structure and trajectory of context sensitivity in LLMs: Content-order decomposition and variance dissociation. *Preprints.org*. <https://doi.org/10.20944/preprints202603.1116.v1>
- Laxman, M. M. (2026h). Encoding fidelity and coherent misalignment in non-English clinical AI. *Preprints.org*. <https://doi.org/10.20944/preprints202604.0061.v1>
- Laxman, M. M. (2026i). Measurement matters: Embedding model choice determines encoding fidelity assessment in multilingual clinical AI. *Zenodo*. <https://doi.org/10.5281/zenodo.19466613>
- Reimers, N. & Gurevych, I. (2020). Making monolingual sentence embeddings multilingual using knowledge distillation. *Proceedings of EMNLP 2020*. arXiv: 2004.09813.
- Feng, F., Yang, Y., Cer, D., Arivazhagan, N., & Wang, W. (2022). Language-agnostic BERT sentence embedding. *Proceedings of ACL 2022*. arXiv: 2007.01852.
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423. DOI: 10.1002/j.1538-7305.1948.tb01338.x.
- Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. arXiv: 1706.03762.
- Brown, T., Mann, B., Ryder, N., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 1877–1901. arXiv: 2005.14165.
- Singhal, K., Azizi, S., Tu, T., et al. (2023). Large language models encode clinical knowledge. *Nature*, 620(7973), 172–180. DOI: 10.1038/s41586-023-06291-2.
- Jin, Y., Chandra, M., Verma, G., Hu, Y., De Choudhury, M., & Kumar, S. (2024). Better to ask in English: Cross-lingual evaluation of large language models for healthcare queries. *Proceedings of the ACM Web Conference 2024*. arXiv: 2310.13132.
- Pfohl, S. R., Cole-Lewis, H., Sayres, R., et al. (2024). A toolbox for surfacing health equity harms and biases in large language models. *Nature Medicine*, 30, 3590–3600. DOI: 10.1038/s41591-024-03258-2.
- Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). Risks from learned optimization in advanced machine learning systems. *arXiv preprint*. arXiv: 1906.01820.

- Shanahan, M. (2024). Talking about large language models. *Communications of the ACM*, 67(2), 68–79. DOI: 10.1145/3624720.
- Huang, L., Yu, W., Ma, W., et al. (2024). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2). arXiv: 2311.05232.
- Conneau, A., Khandelwal, K., Goyal, N., et al. (2020). Unsupervised cross-lingual representation learning at scale. *Proceedings of ACL 2020*, 8440–8451. arXiv: 1911.02116.
- Blasi, D. E., Henrich, J., Adamou, E., Kemmerer, D., & Majid, A. (2022). Over-reliance on English hinders cognitive science. *Trends in Cognitive Sciences*, 26(12), 1153–1170. DOI: 10.1016/j.tics.2022.09.015.
- Kaplan, J., McCandlish, S., Henighan, T., et al. (2020). Scaling laws for neural language models. *arXiv preprint*. arXiv: 2001.08361.
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12, 157–173. DOI: 10.1162/tacl_a_00638.
- Laban, P., Hayashi, H., Zhou, Y., & Neville, J. (2025). LLMs get lost in multi-turn conversation. *arXiv preprint*. arXiv: 2505.06120.
- Tishby, N. & Zaslavsky, N. (2015). Deep learning and the information bottleneck principle. *2015 IEEE Information Theory Workshop (ITW)*, 1–5. DOI: 10.1109/ITW.2015.7133169.
- Fierro, C., Dhar, R., Stamatiou, F., Garneau, N., & Søgaard, A. (2024). Defining knowledge: Bridging epistemology and large language models. *Proceedings of EMNLP 2024*, 16096–16111. DOI: 10.18653/v1/2024.emnlp-main.900.
- Kuratov, Y., Bulatov, A., Anokhin, P., Rodkin, I., Sorokin, D., Sorokin, A., & Burtsev, M. (2024). BABILong: Testing the limits of LLMs with long context reasoning-in-a-haystack. *NeurIPS 2024 Datasets and Benchmarks Track*. arXiv: 2406.10149.